

A certified abstention band for card-fraud decisions, and two quantum arms reported as measured

Global Quantum + AI Challenge 2026 — HSBC track

1 The decision, and what is missing from it

An issuer authorising a card transaction has three actions, not two: approve, decline, or challenge the cardholder through 3-D Secure. The middle action is what makes the problem tractable at scale — routing even one percent of authorisation volume to human analysts is not operationally possible, while a step-up challenge is — and it is the action that published fraud-detection work almost never certifies. Models are reported by AUC or average precision on a held-out fold. Neither is a statement about a decision, and neither carries a finite-sample guarantee.

What we contribute. A three-valued rule whose *abstention band* carries a distribution-free, finite-sample risk certificate, evaluated under a pre-registration frozen before any model was fitted; and an honest measurement of where two quantum approaches sit against a tuned classical baseline on the same decision.

Both quantum arms returned negative results. A fidelity quantum kernel was rejected by its own a-priori screens before it was run on the task. A matrix-product-state classifier ran and did not beat gradient boosting. We report both in the body rather than the appendix, because a screen that only ever passes is not a screen, and because the pre-registration committed us to reporting whatever it returned.

What we do not claim. No quantum advantage of any kind. No portfolio-level PSD2 compliance figure — the SCA-RTS ceilings govern exemption eligibility across a whole portfolio, not a decline threshold on a fraud-enriched research benchmark, and treating them as a threshold selector is a category error we made and withdrew. No claim that the certificate binds the *unconditional* false-decline rate: it does not, and Section 2 says why that distinction is the point rather than a caveat.

2 Method

The rule. With a classical scorer f and thresholds $\tau_{lo} < \tau_{hi}$ fixed in advance,

$$D(x) = \begin{cases} \text{DECLINE} & f(x) \geq \tau_{hi} \\ \text{STEP-UP, then } g(x) \geq \lambda \Rightarrow \text{DECLINE} & \tau_{lo} < f(x) < \tau_{hi} \\ \text{APPROVE} & f(x) \leq \tau_{lo} \end{cases}$$

The band is $B = \{x : \tau_{lo} < f(x) < \tau_{hi}\}$ and g is the in-band re-scorer, which is where a quantum model would enter if one were usable.

The estimand is band-conditional, deliberately. We certify

$$R(\lambda) = \mathbb{P}(D(X) = \text{DECLINE} \mid Y = 0, X \in B),$$

not the marginal false-decline rate. The marginal rate is dominated by τ_{hi} and would be almost unchanged if g were replaced by a coin flip. A certificate that does not bind the component it is said to license is not evidence about that component — so the harder, smaller-sample quantity is the one we bound, and the cost of that choice is reported in Section 3 rather than hidden.

Four blocks, and two distinct breaks of exchangeability. $\mathcal{D}_{\text{train}}$ (60%) fits f ; $\mathcal{D}_{\text{band}}$ (10%) is the *only* data that may set the band edges, g 's hyperparameters or any screen threshold; $\mathcal{D}_{\text{cal}}$ (10%) certifies the composite rule; $\mathcal{D}_{\text{test}}$ (20%) is read once. Blocks are contiguous and snapped to day boundaries.

Two breaks must not be conflated. The **temporal** break is that $\mathcal{D}_{\text{cal}}$ precedes $\mathcal{D}_{\text{test}}$. The **selective** break is that conditioning on band membership is conditioning on a data-dependent event — if the edges were estimated on the same data used to certify. Freezing the band on $\mathcal{D}_{\text{band}}$ removes the second by construction: membership in B becomes a fixed measurable predicate, and filtering a sequence by a fixed predicate preserves exchangeability of what remains. Only the temporal break survives, and we measure it.

Risk control. Learn-then-Test controls family-wise error over a finite grid fixed in advance, using Hoeffding–Bentkus p -values with Holm. Two risks are controlled jointly: $R(\lambda)$ at level α , and the false-negative rate at α_{FN} .

Table 1: False-decline coverage on $\mathcal{D}_{\text{test}}$, calibrated on $\mathcal{D}_{\text{cal}}$ and judged against the exact Beta-Binomial law rather than against α . The stratified and card-disjoint control arms sit inside the interval at every level; the temporal arm does not.

Arm	α	Empirical	Ratio	Errors	Verdict (interval)
temporal	0.010	0.01494	1.494	1667	breached ([973, 1266])
temporal	0.005	0.00746	1.493	833	breached ([458, 665])
temporal	0.002	0.00264	1.322	295	breached ([161, 292])
temporal	0.001	0.00098	0.977	109	inside ([69, 162])
stratified	0.010	0.00952	0.952	1085	inside ([993, 1292])
stratified	0.001	0.00116	1.158	132	inside ([70, 164])
card-disjoint	0.010	0.00896	0.896	990	inside ([960, 1256])
card-disjoint	0.001	0.00076	0.760	84	inside ([67, 161])

Grey bands are the exact Beta-Binomial interval; red points fall outside it

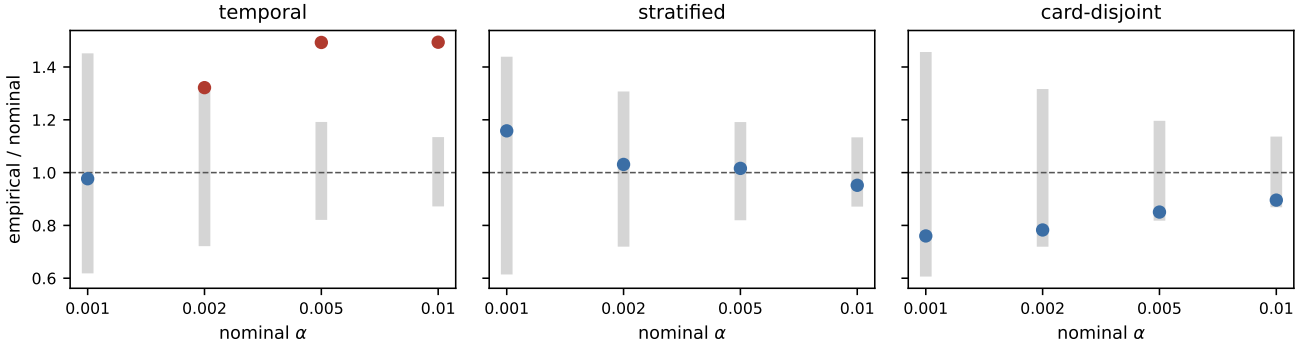


Figure 1: The same data as Table 1, with the Beta-Binomial interval drawn to scale. The interval widens sharply as α tightens, which is why the temporal arm’s apparent compliance at $\alpha = 0.001$ is weak evidence rather than a pass: at that level the law admits almost anything.

The bound applies only to a *mean of per-observation losses*. Our first formulation used $\max(0, \text{floor} - \text{recall}) / \text{floor}$, a nonlinear transform of a mean; the quantities it produced were not valid p -values and the certificate resting on them was not a certificate. It is now a genuine 0/1 loss averaged over frauds, which is equivalent to a recall floor at $1 - \alpha_{\text{FN}}$.

Coverage is judged against the exact law, not against α . For a sound split-conformal system the test-error count follows $E \sim \text{BetaBinomial}(m, n+1-k, k)$ with $k = \lceil (n+1)(1 - \alpha) \rceil$. Checking $\hat{r} \leq \alpha$ is a one-sided test against the wrong null: on a correctly calibrated system it passes only about half the time. Every verdict below is a two-sided Beta-Binomial interval.

Intervals resample entities, not rows. The IEEE-CIS label propagates a reported chargeback to later transactions sharing an account, email or billing address, so rows are clustered: 0.8496 of test-block **card1** values also appear in training. Every interval is a card-level block bootstrap.

3 Results

Coverage holds under two control splits and fails under time. Table 1 calibrates on $\mathcal{D}_{\text{cal}}$ and reports on $\mathcal{D}_{\text{test}}$. The stratified and card-disjoint arms sit inside the Beta-Binomial interval at every α . The temporal arm breaches at three of four levels.

The three-arm design exists to decompose degradation into a temporal component and a sampling component. It does: the controls pass, so the breach is temporal.

But the breach is origin-dependent, and this narrows the claim. Re-running at five rolling calibration origins gives Table 2: the ratio ranges from 0.684 to 1.411. Two of five breach, and the earliest is *conservative* — the deviation reverses direction. So the defensible statement is not that temporal splits inflate the false-decline rate by a fixed factor; it is that the deviation depends on where the calibration window falls, and a single origin cannot

Table 2: The temporal deviation across five rolling calibration origins. Two of five breach and the earliest is conservative, so no single ratio characterises the effect.

Origin (day)	Ratio	Errors	Verdict (interval)
101	0.684	379	conservative ([472, 641])
111	0.893	459	inside ([437, 595])
121	1.411	777	breached ([468, 638])
131	1.198	696	breached ([494, 674])
141	1.126	607	inside ([457, 625])

Table 3: H4 in the band: matrix-product-state against the tuned gradient-boosted baseline on identical rows and features. Intervals are 95 % card-clustered bootstraps. Every interval contains zero, and the comparison is underpowered against its pre-registered ceiling, so what is established is a non-superiority bound rather than a null.

χ	AP (MPS)	AP (GBDT)	Δ AP	95 % CI	p
4	0.1202	0.1407	-0.0205	[-0.0593, +0.0149]	0.864
8	0.1173	0.1407	-0.0234	[-0.0630, +0.0122]	0.895
16	0.1183	0.1407	-0.0225	[-0.0613, +0.0136]	0.884
32	0.1235	0.1407	-0.0172	[-0.0582, +0.0200]	0.806

establish its size. An earlier draft of this work quoted a single seed’s maximum as a central estimate. That was an overclaim, and correcting it removed our cleanest headline number.

What the certificate actually reaches. 5 of 48 configurations certify, all strictly inside the band — the smallest margin below τ_{hi} is 0.0136. That property is not automatic: an earlier version certified a larger set, every one of them sitting exactly at the band edge, where the flagged set is empty by construction and the risk is structurally zero. The certificate was vacuous and we did not notice until we asked where the selected λ sat.

The surviving configurations need $\alpha \geq 0.10$ and $\alpha_{FN} = 0.45$. Nothing certifies at the tightest band budget at all. The mechanism is sample size, not method: at a 2 % traffic budget $\mathcal{D}_{cal}$ contributes only 847 legitimate in-band rows against 58343 overall, and the class-conditional degeneracy floor $\lceil 1/\alpha \rceil - 1$ then binds. We pre-registered this limitation as an amendment before running the arm, and the measurement landed where the amendment predicted.

4 Two quantum arms, both negative

The kernel never reached the task. Before running a fidelity quantum kernel we pre-registered two gates over 120 configurations (encoding $\times$ qubits $\times$ bandwidth $\times$ entanglement). A kernel is usable only if it is not exponentially concentrated — effective rank $r_{eff} = (\sum_i \sigma_i)^2 / (n \sum_i \sigma_i^2)$ inside a usable band — *and* not reproducible by a tuned RBF, $\rho_{RBF} < 0.60$.

28 of 120 pass conditioning. **0 pass both.** The closest any configuration came to distinctness was $\rho_{RBF} = 0.6291$. Every kernel we could construct on this data is, to within measurement, a classical RBF in disguise — which is what the dequantisation literature predicts, and which we would not have known had we reported a downstream accuracy instead.

The closest published comparator agrees. A controlled benchmark of quantum kernel methods on real card-transaction data (Faryad, arXiv:2608.15718, August 2026) reports no robust quantum advantage, with all effect sizes below 0.013 ARI, the sign of the difference dependent on register size, and additional qubits degrading performance through kernel concentration. Its methodological finding is the more useful one for us: a search-budget ablation turned that paper’s one significant advantage into a procedural artefact. Matching search budgets and fixing the screens in advance is what stops the same thing happening here.

The tensor network ran, and did not win. A matrix-product-state classifier contracts $f(\mathbf{x}) = \sum_{\{s\}} A^{s_1} \dots A^{s_N} \prod_j \phi^{s_j}(x_j)$ with $\phi(x) = [\cos \frac{\pi x}{2}, \sin \frac{\pi x}{2}]$, sweeping the bond dimension χ rather than tuning it. In the band, on identical rows and features, it reaches average precision 0.1235 at the largest bond dimension against 0.1407 for the tuned gradient-boosted baseline, a paired difference of -0.0172 . Table 3 gives the sweep.

At full scale it loses by a wider margin. On all 431 features, trained on $\mathcal{D}_{train}$ and scored once on $\mathcal{D}_{test}$, the tensor network reaches 0.8074 ROC AUC and 0.2464 average precision at best across the sweep, against

Table 4: The tensor network at full scale: all 431 features, no band, trained on $\mathcal{D}_{\text{train}}$ and scored once on $\mathcal{D}_{\text{test}}$. The tuned gradient-boosted baseline reaches 0.8837–0.8884 ROC AUC and 0.5055–0.5114 average precision on the same split. Fit time is flat across the bond dimension because a 431-site chain is bound by the launch overhead of its sequential contractions, not by their arithmetic.

χ	ROC AUC	AP	Final loss	Fit (s)
4	0.8074	0.2149	0.3456	2765
8	0.7989	0.1789	0.3075	3071
16	0.7983	0.2464	0.3120	2840
32	0.7889	0.1693	0.3652	2898

0.8837–0.8884 and 0.5055–0.5114 for the tuned baseline. The gap is roughly a factor of two on average precision, which is the metric that matters at a 3.412% positive rate.

That comparison is affordable only because a matrix product state has no qubit ceiling, and because its cost turns out not to depend on the bond dimension the way the ansatz suggests: the four jobs took 2765 to 3071 seconds against the sixty-four-fold spread a χ^2 model predicts. A 431-site chain is bound by the launch overhead of its sequential contractions, not by their arithmetic, so capacity is nearly free and depth is the cost.

The gate meant to tell us whether that was informative was itself wrong. The pre-registration requires the minimum detectable effect to be computed *before* the comparison, and commits to declaring the test underpowered if it exceeds 0.023 average precision. Two defects followed. First, the calculation had no caller at all, so the comparison would have been *observed* rather than tested; the script now refuses to start without it. Second, our noise estimate was the seed-to-seed spread of the baseline against itself, standard error 0.0025. But this comparison is between model *families*, whose measured standard error is 0.0199 — 0.0164 once measured against a second classical family — roughly eightfold larger — so the minimum detectable effect this comparison actually supports is 0.0461, which *exceeds* the pre-registered ceiling of 0.023.

H4 is therefore reported as underpowered, which is what the protocol commits to when this happens. What survives is weaker than a null result and we state it as such: every interval contains zero, so any improvement from the tensor network is bounded above by +0.02 average precision. That is a non-superiority bound. It is not evidence of equivalence, and it is not evidence that the tensor network is worse.

5 Limitations, and what would change the answer

The population is conditional on an incumbent. IEEE-CIS contains only transactions an existing stack already approved; declined attempts cannot generate a chargeback. The estimand is defined on the approved-and-settled population. That is exactly the population a champion/challenger comparison runs on, so the restriction is the right one — but it is a restriction, and no result here extrapolates past it.

The label is partly an entity flag. A chargeback propagates to later transactions sharing an account, email or billing address. Many $Y = 1$ rows are not fraudulent transactions but transactions on a card that later had one. Fraud detection on this file is partly entity-contamination detection.

The certificate is sample-starved by design. Conditioning on a 2% band is what makes the guarantee bind the component we care about, and it is also what leaves 847 legitimate rows to certify with. A larger $\mathcal{D}_{\text{cal}}$, or a wider band, buys tighter α — nothing about the method needs to change.

The tensor-network comparison is narrower than it looks. Published work applying matrix product states to tabular data is largely *generative* — synthetic-data modelling benchmarked on fidelity and privacy. We found no directly comparable discriminative benchmark against a tuned gradient-boosted baseline on imbalanced payment data, which is why this arm was run; but "we did not find one" is a weaker statement than "none exists", and the body claims only the former.

What would change the quantum conclusion. The screens rejected every kernel we could build because each was reproducible by a tuned RBF. An encoding whose Gram matrix is demonstrably outside the RBF family, at a bandwidth that does not concentrate, would pass the same gates unmodified — we did not find one. For the tensor network, the honest requirement is a comparison powered for cross-family noise: at the measured standard error, resolving an effect the size of the interval above needs roughly eight times the in-band evaluation set, which a wider band or a longer observation window would supply.

Reproduction. make smoke && make reproduce && make check. The pre-registration is enforced in code, not by discipline: a gate hashes every guarantee-bearing parameter and refuses undocumented change, the test-fold loader persists an access counter and raises on a second configuration, and every number in this document is generated from a results table rather than typed. The decision log records all thirty-one entries, including the four

retractions above.